NGUYEN TIEN

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PROFESSIONAL SUMMARY

Third-year student pursuing an **Engineer's Degree (Master's level equivalent) in Information Technology (Japanese)** at **Danang University of Science and Technology (DUT)**. Completed an **AI Engineer internship at Vingroup (VinUniversity × VinSpace), 1 of 3 selected from 1,000 AI in Action Cohort 3 trainees** for VinSpace, delivering a star tracker and Moon detection software for a CubeSat 8U. **Codeforces Expert (1731)**. Satellite vision, AI infrastructure, on-premise document AI, and guarded LLM systems. Japanese JLPT N4 (pursuing N3).

EDUCATION

Danang University of Science and Technology (DUT) — Danang, Vietnam
Engineer's Degree (Master's level equivalent), Information Technology (Japanese) – **GPA: 3.33 / 4.0** — *Aug 2024 – Sep 2029*

- **Honors Admission:** Admitted to 3 top-tier tech universities (DUT, VNU-UET, VKU); member of DUT's Informatics Olympiad & ICPC teams.

VinUniversity – *Professional Certificate, AI in Action (Practical Artificial Intelligence)* — *Jul 2026 – Oct 2026*

TECHNICAL SKILLS

- **Programming & DSA:** C/C++ (competitive, advanced), Python, Java, C#, JavaScript/TypeScript, SQL, Lua, algorithms & optimization.
- **AI, CV & Agents:** OpenCV, YOLOv8, ONNX Runtime, sub-pixel centroiding, star identification, PyTorch, QLoRA, LangGraph, multi-agent (MCP, A2A), RAG, tool calling, Document AI / OCR (EasyOCR, VietOCR), Ollama, Gemini.
- **AI Infra & MLOps:** NVIDIA Jetson, Docker, Terraform, AWS, FastAPI, llama.cpp (TTFT/P99), MLflow, DVC, GitHub Actions, Airflow, dbt, Kafka, Delta Lake, DuckDB, Great Expectations, Feast, Qdrant, pgvector, Chroma, FAISS, Redis, HA/DR, Prometheus/Grafana, GPU FinOps.
- **LLMOps & Safety:** Langfuse, LangSmith, RAGAS, OpenTelemetry, Vietnamese PII redaction (regex), prompt-injection guardrails.
- **Full-stack & DB:** React, Next.js, Node.js/Express, PostgreSQL, Supabase (Auth, Realtime, RLS), Tailwind CSS, SQLAlchemy.
- **Languages:** English (intermediate; comfortable reading and listening), Japanese (JLPT N4, actively pursuing N3).

EXPERIENCE & LEADERSHIP

Vingroup (VinUniversity × VinSpace) – AI Engineer Intern — Jul 2026 – Oct 2026

VinSpace (Sep–Oct 2026, selected as 1 of 3 from 1,000 Cohort 3 trainees): onboard vision software for a CubeSat 8U targeting Jetson Orin Nano

- Solved **96.7% of 2,304 simulated flight-camera frames within 10"** with **zero mis-points ≥1°** (ESA tetra3: 86.0%; LOST: 76 mis-points) by building a no-ML lost-in-space star tracker with K-vector Pyramid star ID, SVD Wahba, and a validation gate.
- Sustained **1 Hz attitude solutions on a CPU-only Jetson Orin Nano** (**3,656/3,656** full-res 24.6 MP frames at ~0.74 s, 1.9 GiB, 6.1 W) by porting the solver to a flight package with ground-built catalogs and a CRC-checked 42-byte packet.
- Solved **6/6 real night-sky photos and 8/8 FLIR lab frames** (LOST misidentified 7/8) by recovering focal scale from star geometry alone.
- Detected the Moon in **91.4% of 220 held-out real images at 0.9% false alarms** (sim-augmented baseline: 83.0% / 4.7%) by curating 437 Moon and 1,162 hard-negative NASA images, splitting by capture session, and retraining YOLOv8n on real plus downscaled-real data.
- Scoped flight-compatible training data, **rejecting 8 of 12 (67%) Earth-image datasets** for wrong orbit, sensor, or GSD and selecting BUPT-1 telemetry at **65% input coverage**, by scoring every public source against the CubeSat camera and channel list.

VinUniversity (Jul–Sep 2026): AI in Action training – 15 theory days and 13 Track 2 (AI Infrastructure) labs

- Flagged **7 multi-agent failure modes** (loops, bottlenecks, redundant calls, prompt injection) by shipping AgentLens, a live RCA platform with a 5-node LangGraph (Triage → Evidence → Analyst → Verifier → Alerter), PII redaction, and human-approved config diffs.
- Helped win **2nd Prize (1st Runner-up)** at the room Mini Hackathon by mining **2,522 chatlog lines from 369 learners** to prove a 53.6% same-concept follow-up rate, the evidence behind the team's grounded AI-tutor feature.

Google Student Ambassador (GSA Vietnam) — National / Danang | 2025 – Jul 2026

Official Ambassador – Central Vietnam Representative

- Earned **Top 200 "Gemini Rising Star"** after selection into the **Top 600 of 7,000+** applicants nationwide by publishing 10+ Gemini explainers and running hands-on Gemini workshops at Swinburne University and DUT.

Technical Communities & Informatics Mentoring — Remote / Hue | 2020 – Present

Creator, Administrator & Competitive Programming Coach

- **Kham Pha Khoa Hoc (2020–2021):** Grew a STEM forum from 60K to **400K+ members** by co-founding it and running 5+ online contests.
- **Mini World VN (2020–2022):** Reached **324,000+ downloads** by programming Lua gameplay maps; also served as Official Platform Critic.
- **Olympiad Coach (2022–Present):** Coached Hue students to **a provincial 1st Prize and a 2nd Prize** in Informatics by teaching C++/DSA.

SELECTED PROJECTS

VietIDP – On-premise Vietnamese Document AI — github.com/Tienes2810/OCR-LLM-RESEARCH

DUT Faculty of IT Scientific Research Initiative – Project Team Lead & Lead AI Developer — *2025 – 2026*

- Earned **Top 7 Honorable Mention** at the faculty research day and cut OCR character error on administrative documents from **35.2%** (raw EasyOCR) **to 23.5%** by building a zero-cloud-egress parser: YOLOv8 stamp removal, VietOCR, 4-bit Qwen2.5-7B, and local RAG.

Japanese Learning Platform (PBL3) – Grade A — kanjilearning.vercel.app

Project Lead & Full-stack Developer – github.com/Thainguyen2103/PBL3 — *2026*

- Earned **Grade A** (also on PBL1 and PBL2) by leading a full-stack Japanese-learning app whose real-time PvP matchmaking locks each waiting room with PostgreSQL `FOR UPDATE SKIP LOCKED`, so concurrent joins never overfill a match.

AWARDS, HONORS & SCHOLARSHIPS

- **Competitive Programming: Codeforces Expert** (Peak: 1731); **ICPC Honorable Mention** (National 2025, Southern 2025, Central 2024); **Silver Medal** at DUT Code League 2025; **Central AI Olympiad Regional** (Top 5 DUT); **VNOI Cup 2023 National Finalist**.
- **High School Contests:** Honorable Mention at Central & Highlands Olympiad (2023) and National Youth Informatics (2023); Provincial Informatics Contest: two 3rd Prizes (2022–2023) and a Consolation Prize, Specialized board (2023).
- **Honors & Scholarships: Exemplary Youth Following Uncle Ho's Teachings** (university level, DUT, 2026); **Google Career Certificates Scholarship** (2026); **Stanford Machine Learning Specialization** (Coursera); **Acecook Happy Scholarship** (2024); **Tiep Suc Den Truong Scholarship** (Tuoi Tre Newspaper, 2024); **Hue Study Encouragement Scholarship** (2025).